

Geometry Practice Midterm

January 13, 2025

Problem 1

Restate the following in statements in an "if α then β " form so that α implies β .

- a) I will only mow the lawn if you pay me twenty dollars.
- b) Everyone who passed the test got an A in the class.
- c) Only the wise can understand this statement.

Problem 2

The only way for Dorothy to get to Oz is to follow the yellow brick road.

- a) What can you conclude if Dorothy got to Oz?
- b) What can you conclude if Dorothy follows the yellow brick road?

Problem 3

If P and Q are points in the interior of $\angle O$, prove that the segment PQ is in the interior of $\angle O$.

Problem 4

Prove that every line has a point on either side of it.

Problem 5

In $\angle AOB$, where $\angle AOB$ is NOT a straight angle, point P lies between A and B . Prove that P is in the interior of the angle.

Problem 6

Prove that if $AB \not\cong CD$ then $AB \neq CD$.

Problem 7

Prove that $0.\overline{9} = 1$.

Problem 8

A line l does not contain any of the vertices of a triangle $\triangle ABC$. The line l intersects side AB , but does NOT intersect side BC . Prove that l must intersect the side AC .

Problem 9

Prove that the base angles of an isosceles triangle are congruent.

Problem 10

Prove that any two right angles are congruent.

Problem 11

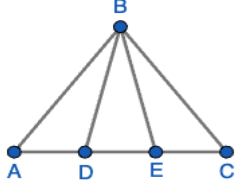
Use Theorem 7-1 to prove that the angles opposite the legs of a right triangle are acute angles.

Problem 12

Prove the following corollary to Theorem 7-1. There is at most one line perpendicular to a given line l from a given point P . [Hint: Suppose that there were two perpendiculars, and derive a contradiction]

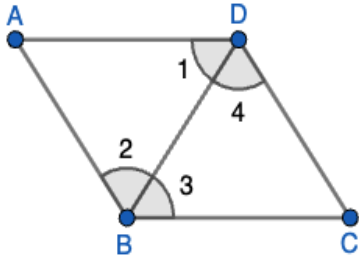
Problem 13

If $\triangle ABC$ is isosceles and $\angle ABD \cong \angle CBE$, prove that $\angle BDE \cong \angle BED$.



Problem 14

If $\angle 1 \cong \angle 3$ and $AD \cong BC$, prove that $\angle 2 \cong \angle 4$.



Problem 15

Determine if each of the following is true or false.

- a) If a theorem is true, its converse is true.
- b) The complement of a 40° angle is a 50° angle.
- c) If segment AB intersects segment CD at point O , then $\angle AOD \cong \angle BOC$.

Problem 16

Prove that the angle sum of any triangle is a straight angle.